NAINIKA GUPTA

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EDUCATION

Duke University **Aug 2026 – May 2030**
B.S.E. in Biomedical Engineering | Certificate in Innovation & Entrepreneurship Durham, NC
- Relevant Coursework: BIO201L, EGR105, EGR101, MATH218D-2
- Activities: Segura Lab (Spring '27)

National University of Singapore (NUS) **Jan 2025 – Aug 2025**
Dual Enrollment Student Researcher Singapore
- Conducted university-level geography research through a highly selective program by the Ministry of Education and NUS
- Authored an independent research thesis on wetland conservation, operating under direct mentorship of university faculty

EXPERIENCE

Agency for Science & Technology Research (Institute of Cellular & Molecular Biology) **Dec 2025 – Jun 2026**
Synthetic Biology Research Intern, Co-Author Singapore
- Investigated CAST genome-engineering systems through saturation mutagenesis across 95 target sites; conducted 1000+ qPCRs and designed saturation mutagenesis libraries to enhance editing efficiency for genetic eye disorders by 6-fold
- Performed a meta-analysis to condense findings, helping secure a SGD100,000 grant for my lab; collaborated with my supervisor through weekly discussions to design the high-throughput screening of 1700+ protein mutations within 3 months

Research Science Institute (RSI) **Jun 2024 – Aug 2024**
Research Scholar Boston, MA
- Selected as 1 of 100 scholars internationally to investigate NETosis regulators in hyperglycemic cardiomyocytes; evaluated Dex-TO as a theranostic platform for myocardial infarction using immunofluorescence assays and murine imaging data
- Named a Top 10 Oral Presenter among RSI scholars, presenting my biomedical research to scientific and industry leaders including the CEO of Regeneron Pharmaceuticals; presentation was broadcast live to the global RSI alumni community

National University of Singapore Nanomaterials Research Lab **Apr 2022 – Jun 2025**
Materials Engineering Researcher Singapore
- Led a research team for 2 years to investigate the focused-laser aggregation of silver nanoparticles on graphene oxide films and zinc oxide nanowires for steganography and chemical sensing; performed characterization with FM, SEM-EDX, Raman
- Independently developed a method to transform rice husk ash into functional carbon-silica microspheres by CNC and pulsed-laser processing; characterized optical imaging behavior and antimicrobial properties; worked ~20h/week for 8 months

LEADERSHIP & AWARDS

Raffles Chess Club **May 2022 – May 2025**
Co-President
- Ranked #5 in Women's Chess in Singapore (Feb 2024); Teplice Intl Open Tournament 2023: Top 8 Women; National and Zonal Team Champions ('20, '21, '22, '23, '24, '25); National Age Group Girls Under 16 & 20 Champion 2022
- Led training across middle and high school divisions and expanding accessibility through level-based training for members
- Organized and arbitrated an inter-school chess tournament for 8 high schools across Singapore; earned National Arbiter norm

youthSTEM **Feb 2024 – Jun 2025**
Co-President & Mentor
- Collaborated with Science Center Singapore and NUS to design and deliver STEM education programs for 50+ elementary and middle school students, featuring hands-on experiments, mentorship and faculty presentations (Prof. Sow Chorng Haur)
- Led a team of 12 people to connect with various local schools, liaise with collaborators, design experiments and plan logistics for STEM camps; planned and spearheaded the training of student volunteers who served as mentors during the camps

Awards:
- **National STEM Talent Search:** Commendation Award (Top 9 / 600+ projects); awarded for excellent independent research
- **Singapore Science & Engineering Fair:** 4x award recipient ('22, '23, '24, '25) for research in materials engineering
- **National Geographic Slingshot Challenge 2025:** Top 0.5% / 2700+ submissions for an original environmental innovation
- **Superintelligence Imagined Creative Contest:** 1 of 6 winners globally; produced an educational video on superintelligence

SKILLS
- **Technical Skills:** Python, Arduino, ANOVA, Experimental Design, Molecular Biology, Nanomaterials Synthesis
- **Languages:** English (Native), Mandarin (Native), Korean (Intermediate), Hindi (Intermediate), Spanish (Beginner)